

BENCHMARK: MARGINAL

Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2458420.6836 +/- 0.0057 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.078 +/- 0.019                 |
| Transit depth (Rp/Rs)^2                           | 0.61 +/- 0.3 [%]                |
| Orbital Inclination (inc)                         | 89.63 +/- 1.43                  |
| Airmass coefficient 1 (a1)                        | 0.9776 +/- 0.0099               |
| Airmass coefficient 2 (a2)                        | 0.017 +/- 0.015                 |
| Scatter in the residuals of the lightcurve fit is | 1.01 %                          |
| Best Comparison Star                              | #2 - [355, 154]                 |
| Optimal Aperture                                  | 4.23                            |
| Optimal Annulus                                   | 9.53                            |
| Transit Duration (day)                            | 0.1494 +/- 0.0035               |

Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R fit vs published     | 0.078 ± 0.019 vs 0.1036 (deviation 1.01σ) |
| Duration fit vs published | 0.1494 d vs 0.1567 d (deviation 1.56σ)    |
| Mid-time O-C              | -22.7 min                                 |
| Depth SNR                 | 3.1                                       |
| Residual scatter          | 1.01 %                                    |

Light curve

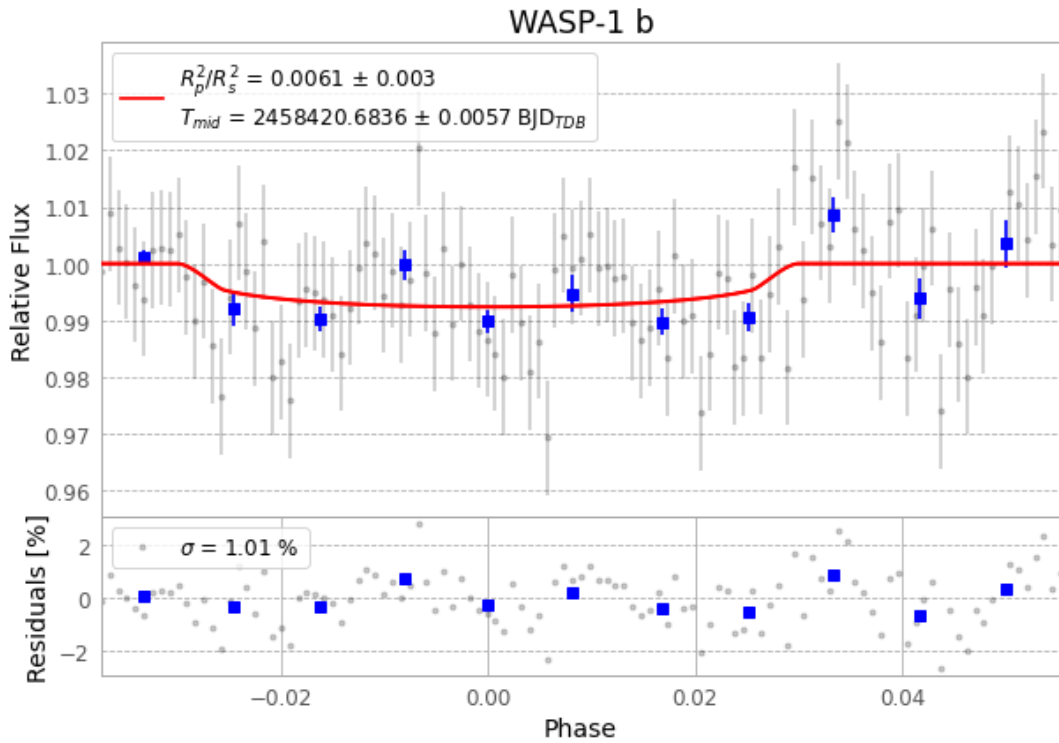


Figure 1 — FinalLightCurve\_WASP-1 b\_2018-10-28.png (SHA-256 0128c69778f92ba9...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [348, 193], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 9b14d2b22d85788f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ba3ef8b

## Data provenance

113 FITS frames (74 MB), WASP-1181029020012.FITS → WASP-1181029073612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-1-181029.log (b6a9739548c4...), FinalLightCurve\_WASP-1 b\_2018-10-28.png (0128c69778f9...), FinalLightCurve\_WASP-1 b\_2018-10-28.csv (c6bb35f24d2f...)

## Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 21:26Z.